

Program : Diploma in Robotic Process Automation	
Course Code : 5309B	Course Title: Non- Relational Database Management Systems Lab
Semester : 5	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Familiarization of No-SQL databases.
- Understand the basic operations using MongoDB.
- Exercise the different functions used in MongoDB.
- To obtain Practical Experience Working with emerging trends.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge of Database Management Systems	3133	Database Management System	3

Course Outcomes:

On completion of the course student will be able to:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Familiarization of No-SQL databases	6	Apply
CO2	Analyze various basic operators in MongoDB	15	Apply
CO3	Use of different functions in MongoDB	12	Apply

CO4	Establish a database connectivity	9	Apply
	Series Test	3	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3	2	2	3	3		
CO2	3	2	2	3	3		
CO3	3	2	2	3	3		
CO4	3	2	2	3	3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Familiarization of No-SQL databases		
M1.01	Familiarization of MongoDB	3	
M1.02	Familiarization of Cassandra	3	
CO2	Analyze various basic operators in MongoDB		
M2.01	Student Database i) Create database, Create collection, Drop collection and database	3	Applying
M2.02	ii) Do the operations, insert data, find, find one, sort	3	Applying
M2.03	iii) Do the operations limit, skip, distinct, projection.	3	Applying
M2.04	Update modifiers (\$set, \$unset, \$inc, \$push, \$pushAll, \$pull, \$pullAll, \$addToSet)	6	Applying

	Series Test-I	1.5	
CO3:	Use of different functions in MongoDB		
M3.01	Indexing in mongodb	3	Applying
M3.02	Mongodb aggregation	3	Applying
M3.03	RDBMS Where Clause Equivalents in MongoDB	3	Applying
M3.04	MongoDB Delete Document	3	Applying
CO4:	Establish a database connectivity		
M4.01	MongoDB with Php i) Make a connection and Select a database ii) Create a collection iii) Insert a document iv) Find all documents v) Update a document vi) Delete a document	9	Applying
	Series Test – II	1.5	

**** - Micro Project**

Students are expected to do a micro project in database management system during the course for the purpose of continuous evaluation. This experiment shall be included in the bona-fide record. Example: Develop program such as

- **Football statistics app**
- **Develop a content Management System**
- **Product Catalog Management**

Text / Reference

T/R	Book Title/Author
R1	Adam Fowler, NoSQL for Dummies, John Wiley & Sons, 2015
R2	NoSQL Data Models: Trends and Challenges (Computer Engineering: Databases and Big Data), Wiley, 2018

Online Resources

Sl.No	Website Link
1	http://mdslab.unime.it/sites/default/files/mongodb_tutorial.pdf
2	https://www.mongodb.com/docs/manual/indexes/
3	https://cassandra.apache.org/_/index.html
4	https://www.projectpro.io/article/mongodb-projects-ideas/640

List of Experiments:

- 1. Familiarization of No SQL databases-Mongodb and Cassandra.**
- 2. Create a student database with the fields: (SRN, Sname, Degree, Sem, CGPA)**
 1. Display all the documents
 2. Display all the students in
 3. Display all the students in ascending order
 4. Display first 5 students
 5. Display students 5,6,7
 6. List the degree of student "Rahul">
 7. Display students details of 5,6,7 in descending order of percentage
 8. Display the number of students in BCA
 9. Display all the degrees without _id
 10. Display all the distinct degrees
 11. Display all the BCA students with CGPA greater than 6, but less than 7.5
 12. Display all the students in BCA and in 6th Sem

3. Create an employee database with the fields: {eid, ename, dept, desig, salary, yoj, address{dno,street, locality, city}}

1. Display all the employees with salary in range (50000, 75000)
2. Display all the employees with desig developer
3. Display the Salary of Rahul
4. Display the city of employee "Rahul"
5. Update the salary of developers by 5000 increment
6. Add field age to employee "Rahul">
7. Remove YOJ from "Rahul"
8. Add an array field project to "Rahul"

9. Add p2 and p3 project to “Rahul”

10. Remove p3 from “Rahul”

11. Add a new embedded object “contacts” with “email” and “phone” as array objects to “Rahul”

12. Add two phone numbers to “Rahul”.

4. Create a student database with the fields: {studentid, Sname, Age, section, subject }

1. Insert multiple documents into the collection

2. Create the index on name field in ascending order.

3. Finding the indexes in a collection

4. Drop indexes in a collection.

5. Display the total number of students in one section only

6. Display the details of students whose age is greater than 30 using match stage

7. Display the distinct names and ages (non-repeating)

8. Display the use of where clause

5. Establish a database connectivity (Mongodb with Php)

- i) Make a connection and Select a database
- ii) Create a collection
- iii) Insert a document
- iv) Find all documents
- v) Update a document
- vi) Delete a document